

1. Business Understanding (DDD — Domain Exploration)

Core business processes

- Customer onboarding and profile maintenance
- Service discovery and selection
- Appointment booking, rescheduling, and cancellation
- Staff assignment based on availability and qualifications
- Check-in, service execution, and appointment completion
- Payment collection, receipt generation, and end-of-day reconciliation
- Customer follow-up, reminders, and rebooking

Main services offered

- Hair services such as cut, wash, styling, coloring, and treatment
- Nail services such as manicure and pedicure
- Beauty services such as makeup, skincare, waxing, or package sessions

Pain points without software

- Paper or chat-based scheduling leads to double booking and lost updates
- Customer history is fragmented, making personalization difficult
- Staff calendars are hard to coordinate, especially during peak hours
- Payment records and daily totals are error-prone when tracked manually
- Reminders are sent inconsistently, increasing no-shows and late arrivals
- Owner has limited visibility into revenue, service popularity, and staff utilization

Goals of the system

- Centralize salon operations in one system
- Reduce booking mistakes and scheduling conflicts
- Improve customer experience with faster and more reliable service
- Support staff coordination and operational transparency
- Automate reminders and routine communication
- Provide reporting for revenue, demand, and performance analysis

2. Functional Features

Feature Area	Description
Customer management	Store customer profiles, contact details, visit history, notes, preferences, and loyalty or discount eligibility.
Appointment scheduling	Create, update, cancel, and reschedule appointments; assign services and staff; prevent overlapping bookings; manage statuses.
Staff management	Maintain employee profiles, skills, roles, working hours, leave days, and service eligibility.
Payments & billing	Record charges, discounts, payment methods, receipts, payment status, and optional refunds or cancellation fees.

Notifications	Send confirmations, reminders, cancellation messages, and schedule change alerts to customers and staff.
Reports	Provide dashboards and reports for appointments, revenue, top services, employee workload, cancellations, and customer retention.

3. Domain Model (Conceptual)

Main entities / aggregates

- Customer
- Employee
- Service
- Appointment
- AppointmentService
- Payment
- Notification
- StaffSchedule

Relationships

- A customer can have many appointments.
- An appointment is assigned to one primary customer.
- An appointment includes one or more services.
- An employee can perform many appointments over time.
- Each service may be offered by one or more qualified employees.
- A payment is linked to one appointment.
- Notifications can be triggered by appointment lifecycle events.

Important business rules

- An appointment must contain at least one service.
- An employee cannot be double-booked for overlapping time ranges.
- Only qualified employees may be assigned to a service.
- Completed appointments can be billed; canceled appointments cannot be completed.
- A payment amount must not exceed the amount due unless refund logic is defined.
- Statuses should follow a controlled lifecycle such as booked → confirmed → completed / canceled.

Value objects

- Money
- TimeRange
- ContactInfo
- ServiceDuration
- Address (optional)

Suggested bounded contexts

- Booking Management — appointments, status changes, service selection, time-slot validation
- Customer Management — customer profiles, contact data, preferences, visit history

- Staff & Availability — employee profiles, skills, schedules, leave, qualification checks
- Billing — pricing, payments, receipts, refunds or fee exceptions
- Communication & Reporting — reminders, alerts, and management dashboards

4. Database Structure (High-Level)

Table	Key fields
customers	id, full_name, phone, email, preferred_contact_method, notes, created_at
employees	id, full_name, role, phone, email, hire_date, status, created_at
services	id, name, category, duration_minutes, base_price, active
employee_services	employee_id, service_id
staff_schedules	id, employee_id, work_date, start_time, end_time, availability_status
appointments	id, customer_id, employee_id, start_at, end_at, status, total_estimate, notes
appointment_services	appointment_id, service_id, quantity, line_price
payments	id, appointment_id, amount, method, status, paid_at, receipt_no
notifications	id, customer_id, appointment_id, type, channel, sent_at, delivery_status

5. User Roles & Permissions

Role	Permissions summary
Owner / Administrator	Full system access; manage employees, services, schedules, appointments, payments, reports, and settings; approve sensitive actions such as refunds or discounts.
Receptionist	Create and edit customers and appointments; assign staff; view payments and reports relevant to front-desk operations; cannot change global settings.
Staff member	View personal schedule; update appointment progress and service notes; view assigned customer details relevant to service delivery; limited edit rights.
Customer (optional)	View own profile and appointments; request or create bookings online; cancel or reschedule within policy limits; view receipts if portal exists.

Permission matrix

Role	View	Create	Edit	Delete	Approve
Owner / Administrator	Yes	Yes	Yes	Yes	Yes
Receptionist	Yes	Yes	Yes	Limited	No
Staff member	Limited	No	Limited	No	No
Customer (optional)	Own data only	Booking request / self-booking	Own bookings within policy	Own bookings within policy	No